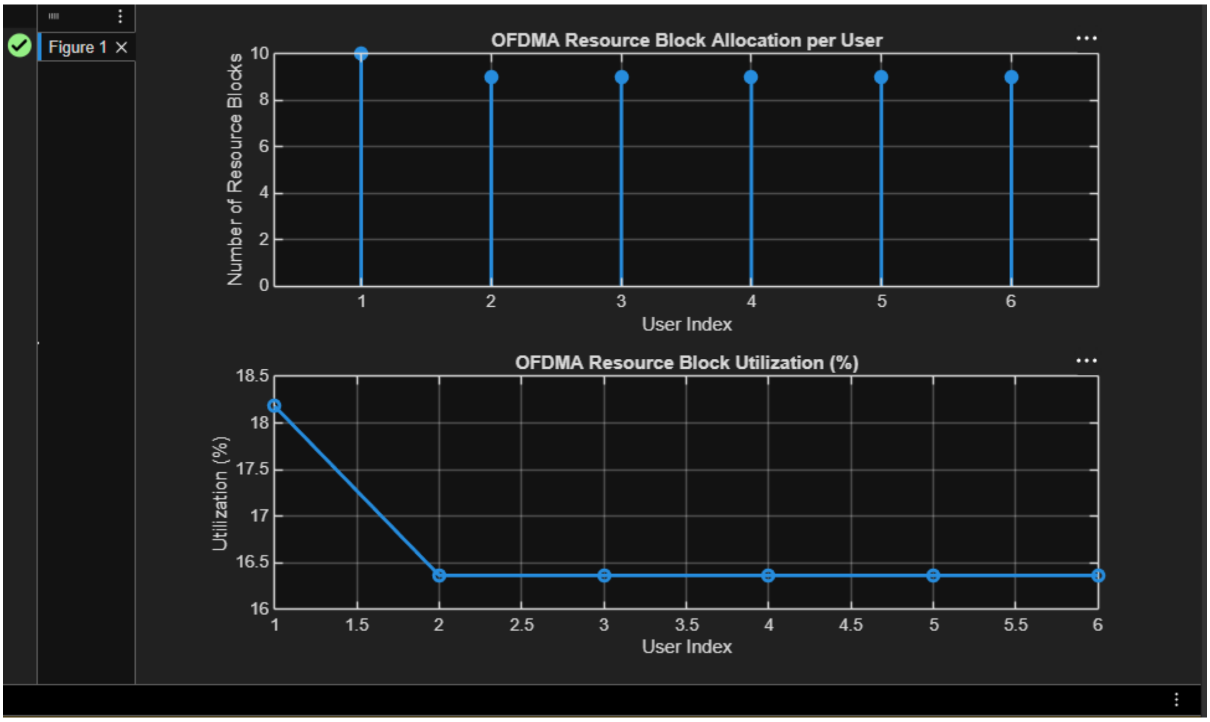
Ex.No.4: OFDMA Resource Block Allocation

Objective 1 Matlab Code and Output:

```
clc; clear; close all;

% System parameters

bandwidth = 10e6;

rb_size = 180e3;

% Total Resource Blocks

% 10 MHz

% 180 kHz per RB

total_RB = floor(bandwidth / rb_size);

% Users

users = 1:6;

% RB allocation (simple equal + remainder distribution)

base = floor(total_RB / length(users));

alloc = base * ones(1,length(users));

alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;

utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')

disp(total_RB)

disp('RB Allocation per User:')

disp(alloc)

figure

% -------- Graph 1: RB Allocation per User --------

subplot(2,1,1)

stem(users, alloc, 'filled', 'LineWidth', 2)

title('OFDMA Resource Block Allocation per User')

xlabel('User Index')

ylabel('Number of Resource Blocks')  grid on
```

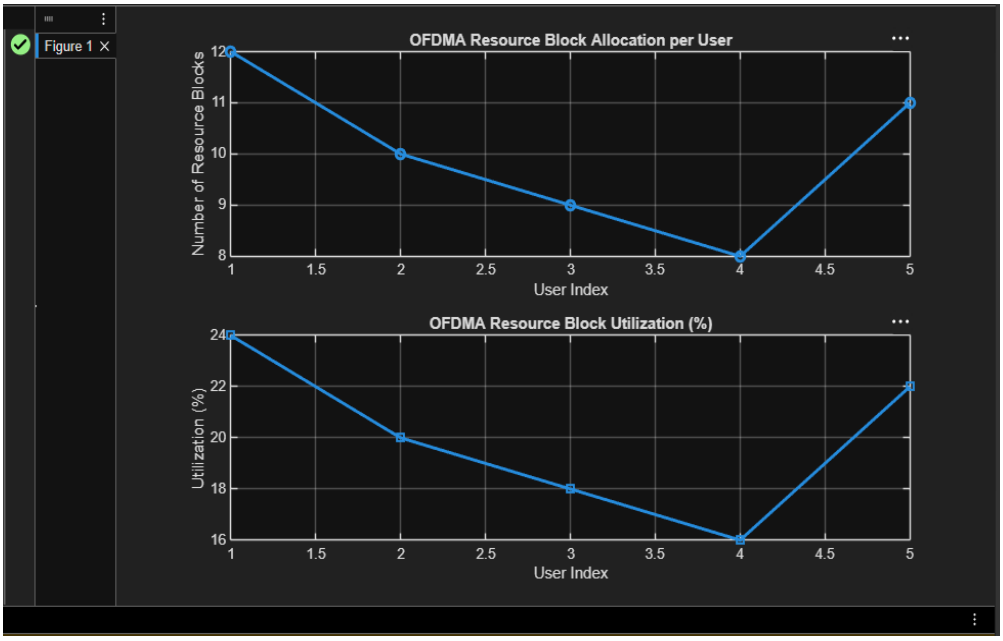
```
% -------- Graph 2: RB Utilization --------

subplot(2,1,2)

plot(users, utilization, '-o', 'LineWidth', 2)

title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')

grid on
```



Objective 2 Matlab Code and Output:

```
clc; clear; close all;

% Total system resource blocks

total_RB = 50;

% Number of active users

users = 1:5;

% RB allocation to users (example scenario)

alloc = [12 10 9 8 11];
```

```
% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;
disp('Resource Block Utilization (%) per User:')
disp(util')
figure
% -------- Graph 1: RB Allocation --------
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% -------- Graph 2: Utilization (%) --------
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```



Objective 3 Matlab Code and Output:

```
clc; clear; close all;

% System parameters

rb_bw = 180e3;

users = 1:5;

% Bandwidth per RB (180 kHz)

% Allocated Resource Blocks per user

alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)

mod_eff = [2 4 6 4 2];  % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

disp('User Throughput (Mbps):')

disp(throughput')

figure

% -------- Graph 1: Throughput per User --------

subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% -------- Graph 2: RB vs Throughput --------

subplot(2,1,2)

plot(alloc_RB, throughput, '-s', 'LineWidth', 2)

title('Resource Blocks vs Throughput Relationship')

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on
```

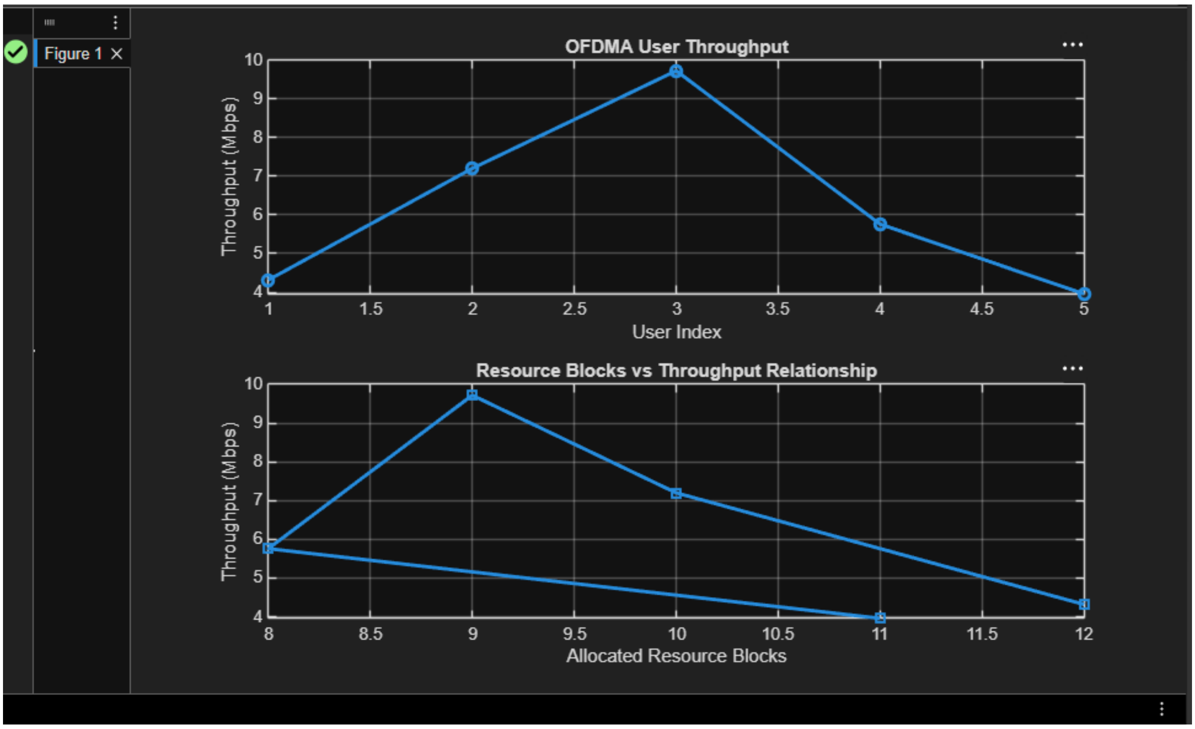


Figure 1
OFDMA User Throughput
Throughput (Mbps)
User Index
Resource Blocks vs Throughput Relationship
Throughput (Mbps)
Allocated Resource Blocks